Tilt-Aware FFT–ICP Scan Registration for Pose-Graph SLAM with a Downward-Tilted Forward-Looking Sonar

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Abstract : A forward-looking sonar (FLS) tilted toward the seafloor breaks the zero-elevation assumption of planar pose-graph simultaneous localization and mapping (SLAM) and biases the scale and overlap of scan registration. We propose a tilt-aware registration that combines fast Fourier transform (FFT) phase correlation with the iterative closest point (ICP) algorithm: it maps both registration inputs to ground-plane range, corrects the beam azimuth of the ICP input, trims ICP correspondences, and reinforces the FFT stage to reduce rejected estimates that fall back on odometry. In a cumulative simulation ablation, ground-plane projection, azimuth correction, and trimming reduced the median trajectory error of the conventional FFT–ICP registration by 79% and 71% on two trajectories (12.86 to 2.65 m and 11.76 to 3.44 m), and the proposed registration cut the odometry fallback rate from 73.6 and 81.8% to 0.4 and 5.4%; the FFT-stage reinforcement reduced fallbacks without a detected accuracy gain, and lost accuracy on Traj. II. A separate back-end trial keeping odometry factors and down-weighting loop closures brought the SLAM median to 0.36 m, overlapping its weak-drift odometry input (median 0.44 m). The evidence is limited to simulation.

1. Introduction

Underwater vehicles operate without GPS, so localization relies on sensors that tolerate turbid, low-visibility water. A forward-looking sonar (FLS) meets that requirement where optical cameras do not. FLS image registration underlies a lineage of pose-graph simultaneous localization and mapping (SLAM) systems [1, 2] that recover trajectories from sonar frames. In previous work [3], a coarse-to-fine registration method first estimated the inter-frame transform by phase correlation computed with the fast Fourier transform (FFT) [4] and then refined it with the iterative closest point (ICP) algorithm. It was evaluated only on frame pairs; SLAM was stated as future work.

One FLS pose-graph SLAM system confines the vehicle state to a plane and sets the sonar's elevation angle to zero because elevation is not recorded in its sonar imagery [5]. A sonar tilted down to observe the seafloor, where no vertical structure is in view [6], breaks that zero-elevation assumption. Prior work has addressed the elevation ambiguity by tracking optical-flow features into a terrain field for a similarly downward-tilted sonar [6], and by fusing the FLS with a second, narrow-beam sonar [7], without modifying the registration step itself. To the best of the authors' knowledge, no prior work quantifies the scale error slant-range projection introduces into FFT–ICP registration, the overlap loss from a body-fixed seafloor-return band, or the effect of either on pose-graph SLAM.

We identify the range mapping, used by both the FFT input image and the ICP point cloud (Fig. 1), as the stage where downward tilt enters registration, and we propose a tilt-aware FFT–ICP registration for pose-graph SLAM (Fig. 2). It maps both registration inputs to ground-plane range, corrects the beam azimuth of the ICP point cloud, trims ICP correspondences, and reinforces the FFT stage to reduce validation-gate rejections that trigger the odometry fallback. The contributions of this study are as follows:

1. **A geometric analysis of two tilt effects.** Under a flat-seafloor model, downward tilt introduces a projection scale factor, reflected in the FFT and ICP scale ratios, and an overlap effect hypothesized to bias the ICP estimate (Sections 3.2 and 3.3).
2. **The proposed registration in pose-graph SLAM.** Ground-plane projection with azimuth correction, trimmed ICP, and FFT-stage reinforcement each target one tilt effect or the odometry fallback, and replace the registration of a planar pose-graph SLAM system (Section 4).
3. **A simulation SLAM evaluation.** In a cumulative clean-odometry ablation (Table 1), ground-plane projection, azimuth correction, and trimming reduced the median absolute trajectory error (ATE) of the conventional FFT–ICP registration [3] by 79% and 71% on two trajectories, and the FFT-stage reinforcement cut odometry fallbacks to 0.4–5.4% with no detected gain and a Traj. II loss (Section 5).

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2. Related Work

FLS registration methods include FFT-based phase correlation [4], sonar-specific feature descriptors [8], and a polar transform that decomposes registration into rotation and translation [9]. Graph matching avoids the local minima of non-convex ICP [10]. The previous FFT–ICP registration underlying this study belongs to this family [3], which aligns sonar images without modeling tilt-dependent range geometry.

The pose-graph SLAM lineage runs from imaging-sonar-aided navigation [1] to factor-graph formulations analyzing the sensor's degeneracies [2]. One representative system uses ICP scan-matching factors and consistency-filtered loop closures on a planar state [5]; this design has been verified only for that system.

Elevation ambiguity has been resolved by optical-flow tracking into a terrain field for a similarly downward-tilted sonar [6], and by fusing a forward-looking sonar with a second sonar [7]. A multibeam acoustic-lens sonar has separately generated three-dimensional point clouds from vehicle motion [11]. Side-scan mapping converts slant to ground range under an altitude-aided flat-floor assumption that sets object altitude to zero, and the resulting error has been quantified [12]; Eq. (1) applies the same correction to FLS registration. Such work recovers height by terrain inference or a second sensor, or quantifies flat-floor error, rather than correcting registration.

3. Pose-Graph SLAM and the Effect of Downward Tilt

3.1. Existing System

The baseline pipeline combines the frame-pair FFT–ICP registration of [3] with a planar pose-graph SLAM back end with the same factor types as [5] (grey blocks in Fig. 2). A smallest-of cell-averaging constant-false-alarm-rate (SOCA-CFAR) detector extracts feature points [5, 13] for the ICP point cloud. Phase correlation

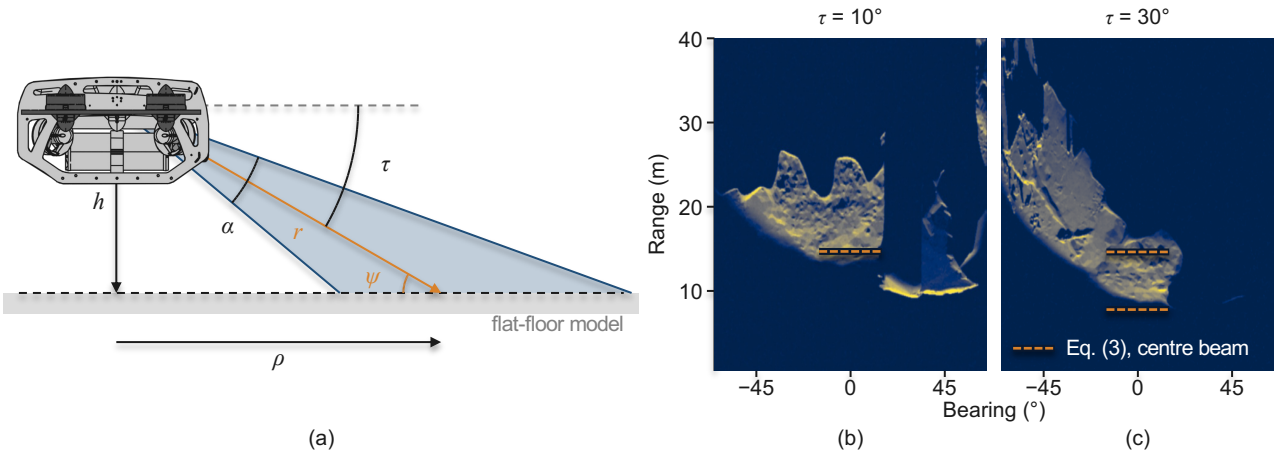


Fig. 1 (a) Downward-tilted forward-looking sonar on a BlueROV2 (schematic, not to scale; τ and α as in the simulation); (b), (c) recorded polar frames of Traj. III and Traj. I at 5.0 m altitude. Dashed lines are the band edges of Eq. (3) under the flat-seafloor model, which the structured simulated floor follows only approximately.

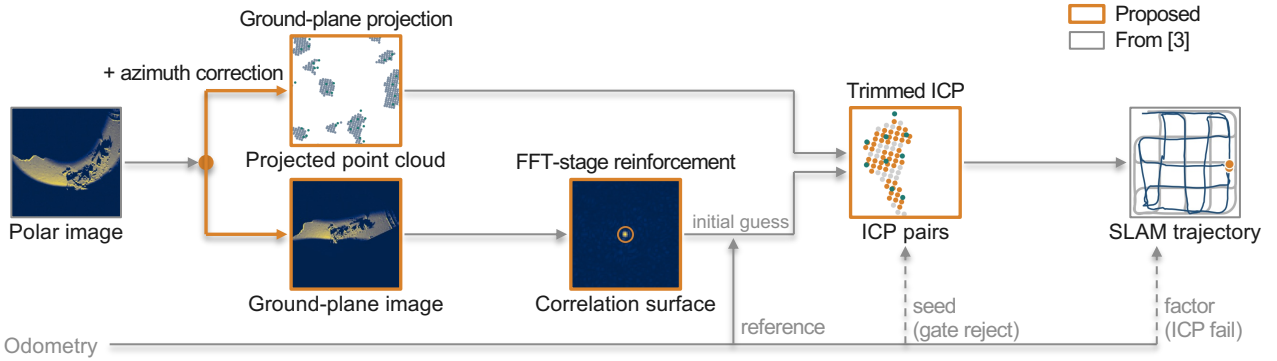


Fig. 2 Overview of the proposed tilt-aware registration in pose-graph SLAM. Highlighted blocks are proposed; grey blocks are the baseline. One range-to-ground projection feeds both the FFT and the ICP stage, and azimuth correction is applied to the point cloud only. For one Traj. I keyframe pair (orange marker on the trajectory), the zoomed ICP panel shows source points kept (orange) and dropped (grey) by trimming ($\gamma = 0.4$) against a target of three accumulated keyframes (cyan).

gives the FFT rotation on the polar image [4] and the translation on a range-mapped image. A fixed gate validates the FFT estimate against the odometry increment before it seeds ICP against recent keyframes [14, 15]; the refined transform forms the sequential factor.

Loop factors come from non-sequential scan matching: candidate keyframes within the field of view are selected by a bounded global search, re-selected by overlap, refined by ICP, checked by pairwise-consistent measurement-set filtering [16], and added with a robust kernel. Incremental smoothing and mapping optimizes the planar pose graph [17]. Odometry validates the FFT estimate, seeds ICP when the gate rejects it, and replaces the registration factor only when ICP initialization or validation fails (Fig. 2).

3.2. Slant-Range Scale Bias

Under a flat-seafloor model (Fig. 1), downward tilt introduces one systematic scale error and one overlap effect into FFT–ICP registration. A sonar return at slant range r maps to a ground-plane range

$$\hat{\rho} = \sqrt{r^2 - h^2}, \tag{1}$$

where h is the altimeter altitude, taken as the sonar's altitude above the floor in level flight, and $\hat{\rho}$ is the flat-floor ground-plane range.

A range mapping $\hat{\rho}(r)$ scales a return's true radial displacement by

$$s = \frac{d\hat{\rho}}{dr} \cdot \frac{\rho}{r}, \tag{2}$$

where $\rho = r\cos\psi$ is the true ground-plane range, ψ is the ray depression angle below horizontal with $\sin\psi = h/r$, and s is the registered-to-true displacement ratio along the beam, per return. Eq. (1) gives $s = 1$ within these assumptions; at $\hat{\rho} = r$, $s = \cos\psi$; at $\hat{\rho} = r\cos\tau$, $s = \cos\tau\cos\psi$, where τ is the tilt below horizontal.

The baseline applies both mappings: the ICP point cloud uses $\hat{\rho} = r$ and the FFT input image uses $\hat{\rho} = r\cos\tau$. Both are the zero-elevation assumption of Section 1, for a level and a τ-tilted sensor plane, and both scale radial displacement down. The FFT translation, scaled by the smaller factor, falls short of odometry by $1 - s$. Once it exceeds the fixed position gate (Section 5.1), the odometry fallback replaces the estimate. The FFT scale ratio of Table 2, taken over all attempted pairs before the gate, reflects this bias in part; the ICP ratio also reflects the azimuth correction (Section 4.1).

3.3. Seafloor-Return Band and Overlap

Under the same flat-seafloor, ideal-aperture model, a seafloor-viewing sonar's returns fall within a slant-range band

$$r \in \left[\frac{h}{\sin(\tau + \alpha/2)}, \frac{h}{\sin(\tau - \alpha/2)}\right], \tag{3}$$

where α is the sonar's vertical aperture and the ray depression angle spans $\psi \in [\tau - \alpha/2,\ \tau + \alpha/2]$; this far edge is finite only when $\tau > \alpha/2$, and otherwise the sonar's range window sets the band instead. This band is fixed to the vehicle body rather than to the seafloor, so the imaged patch shifts forward with the keyframe-to-keyframe

displacement instead of remaining centered on a common footprint. If the returns are assumed to be spread evenly across the band, and cross-track and terrain-relief effects are neglected, the overlap between two keyframes shrinks as their along-track displacement grows relative to the band's along-track ground length. Points falling outside the shared overlap enter spurious nearest-neighbor correspondences, which can bias the recovered ICP translation.

The baseline's ICP keeps a fraction $\gamma = 0.8$ of the correspondences and can therefore retain points outside the shared overlap.

4. Proposed Tilt-Aware Registration

We propose three components that change the registration front end and leave the back end of Section 3.1 unchanged (highlighted blocks in Fig. 2). Ground-plane projection with azimuth correction targets the slant-range scale bias of Section 3.2. Trimmed ICP targets the overlap effect of Section 3.3. FFT-stage reinforcement targets the validation-gate rejections that trigger the odometry fallback.

4.1. Ground-Plane Projection with Azimuth Correction

We apply Eq. (1), using the altimeter reading (the FFT image reuses its mapping until the altitude crosses a 0.25 m step), to both the FFT input image and the ICP point cloud, replacing the two baseline range mappings of Section 3.2 with one shared convention. A beam-azimuth correction restores the horizontal bearing of returns seen by a tilted beam, since the ICP point cloud is otherwise drawn with horizontal-plane bearings. A floor point (X, Y, h) appears at beam azimuth $\beta = \operatorname{atan2}(Y,\, X\cos\tau + h\sin\tau)$; its closed-form inverse is the horizontal bearing $\phi = \arcsin\!\left(c/\sqrt{1+b^2}\right) + \arctan b$, $b = \tan\beta\cos\tau$, $c = \tan\beta\, h\sin\tau/\hat{\rho}$, with the configured tilt τ, altimeter reading h, and $\hat{\rho}$ of Eq. (1); without altitude, no correction applies. It has no free parameter. We apply it to the ICP point cloud only, as it hurt FFT phase correlation in a preliminary trial; +GP adds it with the projection (Table 1).

4.2. Trimmed ICP

We compute the ICP estimate with a trimmed objective

$$E(\mathbf{T}) = \sum_{i=1}^{\lfloor \gamma N \rfloor} \left\| \mathbf{T}(\mathbf{p}_{(i)}) - \mathbf{q}_{(i)} \right\|^2, \tag{4}$$

where $\mathbf{T} \in SE(2)$ is the planar rigid transform being estimated, $\{(\mathbf{p}_i, \mathbf{q}_i)\}_{i=1}^{N}$ are the N correspondences found within the search radius, ordered by increasing residual so that $(\cdot)_{(i)}$ denotes the i-th smallest, and γ is the kept fraction, following the trimmed-objective form of [18]. We fix γ rather than estimate it. A maximum-distance outlier gate is applied jointly with trimming [15], so the kept share is at most γ; the search radius and this gate are held at the same values in every configuration (Section 5.1). Trimming tolerates the partial overlap of Section 3.3; +Trim lowers γ from 0.8 to 0.4.

4.3. FFT-Stage Reinforcement

We reinforce the FFT stage with four measures, added one at a time as configurations (Table 1). (i) +MultiRot: among K rotation hypotheses extracted from the polar-image phase correlation, we carry each candidate to translation and keep the one with the strongest translation-correlation peak, because a single global rotation peak of the polar correlation surface was often wrong at the working tilt. (ii) +Filter: we apply a low-pass mask, switch contrast-limited adaptive histogram equalization (CLAHE) off, and add a Hann window to the translation cross-power spectrum, all three together. The mask suppresses the high-frequency speckle that makes up a large share of that spectrum at the working tilt; CLAHE adds noise to correlations of mostly blank images; the window limits leakage at the zero-padded border. (iii) +Rewarp: for a pair the validation gate rejects, one inverse-warp re-estimation re-aligns the polar images using the rejected estimate and repeats the phase correlation, to recover some rejected pairs. (iv) Proposed: sub-pixel peak localization uses a parabolic fit in place of upsampled discrete Fourier transform (DFT) refinement. Proposed thus has all three components.

Config.	Adds
Baseline	registration settings of [3] ($\gamma = 0.8$, $K = 1$)
+GP	ground-plane projection and azimuth correction
+Trim	trimmed ICP, $\gamma = 0.4$
+MultiRot	$K = 9$ rotation hypotheses
+Filter	low-pass mask, CLAHE off, Hann window
+Rewarp	inverse-warp re-estimation
Proposed	parabolic sub-pixel peak

Table 1 Configurations of the cumulative ablation; each row adds one step to the previous row.

5. Experiments

5.1. Setup and Metrics

Runs used Stonefish [19], a BlueROV2 model, and a 512-beam sonar with $130° \times 20°$ aperture and 0.5–40 m range window. The simulated floor is a static rock-and-structure mesh; its height under the vehicle spans about 5 m (5th–95th percentile), comparable to the 4.6–6.4 m median altitude, so the flat-seafloor model of Section 3.2 holds only approximately. Recorded lawnmower trajectories are Traj. I (640 s) and Traj. II (336 s), both at 30° tilt, and Traj. III at 10°. Seven configurations, Baseline to Proposed (Table 1, Fig. 2), each add one step; Baseline uses the registration settings of the previous study [3] on the present back end and is no reproduction. The search radius (10.0 m), outlier gate (3.0 m), keyframe rule, validation gate, pairwise-consistency set size, ICP convergence threshold, and planar iSAM2 were fixed across configurations, Baseline included. The gate passes an FFT estimate within 0.25 m and 10° of the odometry increment, a keyframe needs at least 1 s and more than 1 m of motion or 10° of rotation, and ICP matches three previous keyframes.

Traj. II informed projection mode, γ, and K (Baseline to +MultiRot) and is held out only for +Filter, +Rewarp, and Proposed, whose settings, the ICP-only azimuth correction, and the pairwise-consistency set size were tuned on Traj. I. Their agreement is consistency, not validation on unseen data.

Clean odometry is the simulator ground-truth pose, so errors are optimistic. Drifting odometry perturbs only the SLAM input, with a 2% (weak) or 10% (strong, author-chosen) translation scale error and a heading random walk of 0.1 or $0.5°/\sqrt{\text{s}}$; evaluation uses clean ground truth. The weak level is about twice the ±1.01% long-term accuracy of a Doppler velocity log (DVL) of this class [20], and its heading error reaches about 2.5° (1σ) at the end of Traj. I, like a calibrated magnetic compass's 1–3° accuracy [21]; since aided heading error is bounded, the walk is conservative. Drift runs used Traj. I with same-queue clean runs; configurations shared noise realizations by round (Table 3).

Metrics. Each configuration ran $n = 4$ times in one queue per trajectory, cyclically shifted between rounds. ATE RMSE pairs each final SLAM keyframe with ground truth at its sensor time before one whole-run SE(2) alignment, the same for SLAM and odometry input. Fallback rate is the share of registrations seeded by odometry after the gate rejects the FFT estimate. FFT and ICP scale ratios are medians of the four run-wise medians of estimated over true translation over attempted pairs (FFT, before the gate) and refinements (ICP); pair sets differ by configuration. ICP converged in every refinement; odometry factors were at most 4.2% of sequential factors (not Proposed†). Cells are median [min, max]; the only (low-power) test is complete separation, i.e., non-overlapping ranges, with exact two-sided $p = 2/70$, or $2/16 = 0.125$, the smallest attainable, between drift configurations paired by round, unadjusted for multiplicity. Pre-declared comparisons were ATE for Baseline vs +Trim and +Trim vs Proposed on Traj. I and II, the fallback rate of Proposed vs +MultiRot, and, under weak drift, Proposed vs clean-odometry Proposed, vs Baseline, and vs +Trim (declared before its added runs; Proposed vs +MultiRot secondary); all other comparisons, the strong-drift level, and the

Config.	ATE RMSE (m)	Fallback (%)	FFT scale	ICP scale
Traj. I				
Baseline	12.86 [12.19, 14.28]	73.6 [73.5, 74.4]	0.719	0.795
+GP	9.50 [8.84, 10.14]	61.1 [61.0, 61.1]	0.918	0.895
+Trim	2.65 [2.10, 3.30]	61.4 [61.1, 61.6]	0.914	0.994
+MultiRot	2.68 [2.21, 2.86]	4.1 [4.0, 4.5]	0.935	0.958
+Filter	2.59 [2.29, 2.81]	2.0 [1.7, 2.0]	0.931	0.958
+Rewarp	2.58 [2.14, 4.67]	0.3 [0.3, 0.3]	0.930	0.950
Proposed	2.35 [2.10, 2.52]	0.4 [0.0, 0.6]	0.932	0.950
Traj. II				
Baseline	11.76 [11.25, 12.37]	81.8 [81.5, 81.8]	0.741	0.766
+GP	11.37 [10.99, 11.53]	61.5 [61.3, 61.5]	0.883	0.833
+Trim	3.44 [3.23, 3.55]	61.5 [61.3, 61.8]	0.883	0.985
+MultiRot	3.29 [3.08, 3.94]	14.4 [14.4, 14.4]	0.913	0.940
+Filter	3.63 [3.11, 3.96]	8.0 [7.0, 10.2]	0.924	0.939
+Rewarp	3.70 [2.97, 3.83]	5.7 [4.3, 5.9]	0.921	0.934
Proposed	4.08 [3.72, 4.30]	5.4 [4.8, 7.0]	0.926	0.936

Table 2 Cumulative ablation on Traj. I and II (median [min, max], $n = 4$; scale ratios: median of the four run medians).

odometry input, declared without a decision rule, are descriptive.

5.2. Contribution of Each Component

Compared with the conventional FFT–ICP registration [3] (Baseline), +Trim reduces the median ATE by 79% and 71% on Traj. I and II with non-overlapping ranges (Table 2, Figs. 3(a) and 4), and the FFT stage adds no detected gain (Proposed 82% and 65%); the larger step is +GP to +Trim. Projection in +GP raises the FFT scale ratio but keeps the trimming of the baseline ($\gamma = 0.8$). Trimming in +Trim drops the largest-residual correspondences and raises the ICP scale ratio to 0.985–0.994, consistent with removing those lost to the body-fixed band (Section 3.3). Descriptively, keyframe-path length over true length rises from Baseline to +GP to +Trim (Traj. I 0.715, 0.842, 0.934; Traj. II 0.773, 0.834, 0.922), though on both +Trim is closer to one than Proposed (0.917 and 0.890). Most of the accuracy gain therefore appears with trimming, whose mechanism is not separated.

Proposed lowers the fallback rate from 61.4 and 61.5% (+Trim) to 0.4 and 5.4%, entirely below +MultiRot on both trajectories, yet the ATE of +Trim and Proposed overlaps on Traj. I. Under clean odometry, a fallback seeds ICP with the simulator's exact motion, so an FFT seed cannot be expected to lower ATE. The FFT-stage gain is therefore fewer fallbacks; its accuracy benefit can show only under drifting odometry (Section 5.3).

On Traj. II, Proposed lies entirely above +Trim (4.08 vs. 3.44 m), possibly because under-scaled FFT seeds replace exact odometry seeds (untested). On Traj. I, scale ratios near one mean scale bias alone does not explain the ATE floor, whose origin is not established. The Traj. II result limits the FFT-stage claim to fewer fallbacks.

5.3. Tilt Angle and Odometry Drift

Proposed lowers the fallback rate of Baseline from 40.4 to 0.9% at 10° (Traj. III), while their ATE overlaps (Table 3, Fig. 3(c)). At this tilt τ does not exceed $\alpha/2$, so under the flat-seafloor model the band has no finite far edge (Section 3.3); in recorded frames it reaches about 26 m at the centre beam (16 m at 30°) (Fig. 1). Traj. III's path differs from Traj. I and II, so the overlap is not attributed to tilt alone.

In the drift queue, Proposed cuts Baseline's median ATE by 80%, 86%, and 75% under clean, weak, and strong odometry (Fig. 3(b)), and its median fallback rate is at most 1.6%, so nearly all ICP refinements start from FFT estimates, not the drifting stream. Under weak drift, +Trim and Proposed overlap (1.93 vs. 1.99 m), as do +MultiRot and Proposed, so no FFT-stage accuracy gain is detected.

The odometry input lies entirely below every default-back-end configuration in Table 3 under weak drift and overlaps Proposed under strong drift. Proposed† (odometry factors kept, loop closures down-weighted) lowered the weak-drift median from 2.39 to 0.36 m (same-noise input 0.44 m), with a setting tuned on Traj. I only.

Config.	ATE RMSE (m)	Fallback (%)
Traj. III, clean		
Baseline	0.88 [0.86, 0.94]	40.4 [40.4, 40.6]
Proposed	0.71 [0.66, 1.21]	0.9 [0.4, 1.3]
Traj. I, clean		
Baseline	12.48 [11.22, 14.88]	73.6 [73.5, 73.6]
+MultiRot	2.44 [2.34, 2.50]	4.8 [4.0, 5.1]
Proposed	2.52 [2.05, 2.64]	0.3 [0.0, 0.9]
Traj. I, weak drift		
Odom. input	0.44 [0.36, 0.65]	n/a
Baseline	13.93 [8.52, 15.90]	78.1 [77.1, 78.2]
+Trim	1.93 [1.55, 3.57]	64.4 [63.6, 65.9]
+MultiRot	2.15 [1.99, 2.41]	4.8 [4.5, 7.1]
Proposed	1.99 [1.77, 2.70]	0.3 [0.0, 0.6]
Proposed†	0.36 [0.30, 0.38]	0.6 [0.0, 1.1]
Traj. I, strong drift		
Odom. input	2.28 [1.85, 3.29]	n/a
Baseline	10.43 [8.60, 12.76]	83.0 [81.4, 83.8]
Proposed	2.62 [2.42, 3.09]	1.6 [0.3, 2.1]

Table 3 Tilt and odometry-drift conditions (median [min, max], $n = 4$). Traj. I clean rows come from the drift run queue, apart from Table 2. †Modified back end, separate queue (its Proposed 2.39 m).

6. Conclusion

Compared with the conventional FFT–ICP registration, ground-plane projection and azimuth correction followed by trimming reduced the median trajectory error of planar pose-graph SLAM by 79% and 71% on two simulated trajectories, and the FFT-stage reinforcement then cut the odometry fallback rate from 61.4 and 61.5% to 0.4 and 5.4% (73.6 and 81.8% for the conventional registration).

Limitations are the simulation-only evidence with clean ground-truth odometry, the unseparated effect of floor-height variation on the scale ratios, a back-end change, tested on Traj. I only, to approach weak-drift odometry, and no detected accuracy gain from the FFT stage, whose full configuration lay above +Trim on partially held-out Traj. II.

Future work covers real DVL and inertial odometry, field data, trajectories unused for any setting choice, seafloor relief varied in a controlled way, and uncertainty-weighted registration factors.

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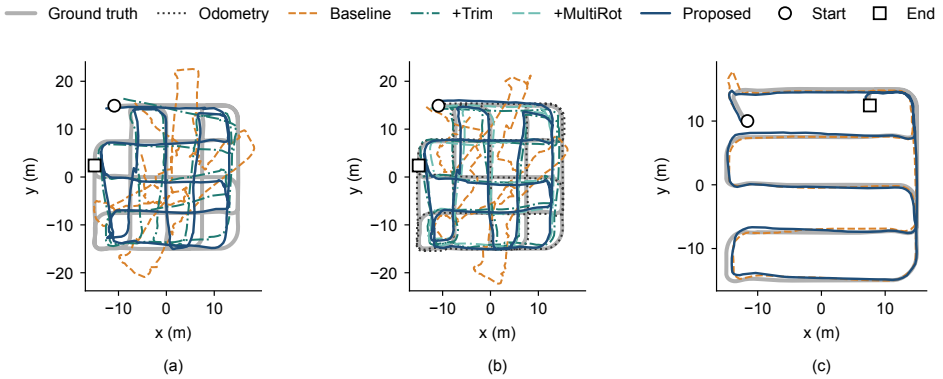


Fig. 3 Top-view trajectories after SE(2) alignment: (a) Traj. I, clean odometry; (b) Traj. I, weak drift; (c) Traj. III ($\tau = 10°$), clean odometry. Each curve is one representative replicate.

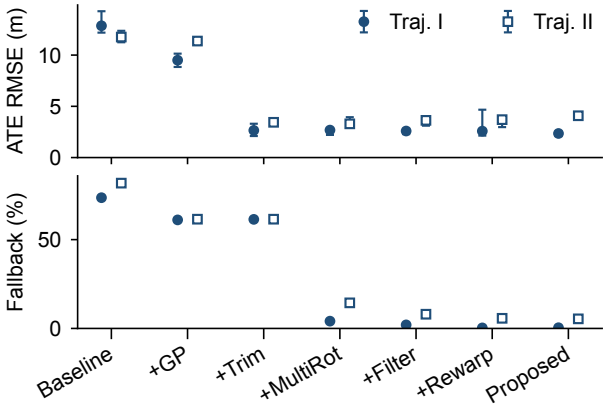


Fig. 4 ATE RMSE (top) and fallback rate (bottom) of the seven configurations on Traj. I and II (median and min–max, $n = 4$).

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